

Customer shipping & sensor onboarding

This guide explains how to provision a customer farm, flash field nodes, and ship hardware so the customer sees their devices in the Persephone's Basket app.

For API details and admin setup, see [backend/README.md](#).

How the pieces fit together

Staff (you)		Customer
Provision farm + N nodes	→	Sign up / log in with email
Print 1 access code on card	→	Enter access code (once)
Flash each node with its id + per-device ingest key		Open app → Sensors tab sees their nodes

Three separate concepts link together:

Piece	Purpose
Farm	The customer's vineyard (tenant boundary in the database)
Access code	Lets a user account join that farm (one code per farm, not per device)
Device id + ingest key	Lets each physical node send sensor data into that farm

The access code and device ingest keys are **independent**:

- The customer gets the **access code** (printed on a card).
- The customer **never** gets ingest keys — those are flashed onto the hardware only.

Example: customer orders 2 devices

Step 1 — Provision the farm (before shipping)

Use the **Admin tab** in the app (staff-only), or the CLI.

In the app: Admin → Provision farm → enter farm name and node count **2**.

CLI (local, with DATABASE_URL):

```
cd communication-platform/backend
npm run provision -- --farm "Smith Vineyard" --nodes 2
```

CLI (on Railway, where the database lives):

```
railway ssh --project=88eca128-a3e4-4094-9c50-3dcec2779c01 \
--environment=f1bf40e0-15ea-456e-a8a0-f52546c24666 \
--service=af43fc9c-0298-40ea-9d72-7adec106f301 \
'cd /app && npm run provision -- --farm "Smith Vineyard" --nodes 2'
```

Provisioning creates:

	Node 1	Node 2
Device id	{farmId}-A1 (e.g. farm_abc123-A1)	{farmId}-A2
Display name	PB Node A1	PB Node A2
Location label	Block 1	Block 2
Ingest key	unique secret (shown once)	unique secret (shown once)

Plus **one access code** for the entire farm, formatted like:

PB-XXXX-XXXX-XXXX-XXXX

Save the provisioning output immediately. Only hashes are stored in the database. Lost secrets must be **reissued**, not recovered.

Step 2 — How devices are named (don't invent your own)

Provisioning assigns names automatically. You do **not** pick arbitrary device ids.

Field	Pattern	Example
Device id	{farmId}-A{n}	farm_abc123-A1
Display name	PB Node A{n}	PB Node A1
Location label	Block {n}	Block 1

Device ids are **globally unique** and namespaced by farm id so two customers never collide (e.g. both having pb-node-A1).

For **N nodes**, ids run A1 ... A{N} (up to 200 per farm).

Step 3 — Flash each node before shipping

Each node must POST readings to `POST /v1/sensors/ingest` with:

Header:

```
x-device-key: <that node's ingest key from provisioning>
```

Body (example for node 1):

```
{
  "deviceId": "farm_abc123-A1",
  "deviceName": "PB Node A1",
  "farmName": "Smith Vineyard",
  "locationLabel": "Block 1",
  "status": "online",
  "readings": [
    { "sensorType": "temperature", "value": 22.5, "unit": "C" },
    { "sensorType": "humidity", "value": 61.0, "unit": "%" },
    { "sensorType": "soil_moisture", "value": 38.0, "unit": "%" }
  ]
}
```

Important:

- Use the **exact** `deviceId` and `x-device-key` from provisioning output.
- `farm_id` is set at provisioning time and **cannot** be changed by the device payload.
- Prefer `x-device-key` (per-device). The shared `SENSOR_INGEST_API_KEY` (`x-sensor-key`) is **legacy only** for nodes already in the field before multi-tenancy.

Step 4 — What goes in the box

Item	Notes
Device 1	Flashed with <code>{farmId}-A1</code> + its ingest key
Device 2	Flashed with <code>{farmId}-A2</code> + its ingest key
One access code card	Same code works for the whole farm
Ingest keys	Never include these in the box — firmware only

You do **not** need two access codes for two devices. **One code = one farm = all devices on that farm.**

Step 5 — What the customer does in the app

1. **Sign up or log in** (email + password).
2. If they belong to no farm yet, the app shows the **Connect** / access-code screen.
3. They enter the code from the card: `PB-XXXX-XXXX-XXXX-XXXX` .
4. The app calls `POST /v1/farms/claim` , which adds them to `farm_members` .
5. The main app opens. On the **Sensors** tab (Demo mode), the app loads `GET /v1/sensors/overview` .
6. The API returns **only devices** whose `farm_id` matches farms the user has joined.

The access code is an **enrollment credential**, not a password:

- It is exchanged **once** for permanent membership.
- It is **not** stored on the phone after redemption.
- Revoking a code later does **not** remove users who already joined.

Step 6 — What they see on the vineyard map (Demo mode)

In **Demo**, live sensor nodes map to map blocks by the **A-number** in the device name/id:

Device	Map block
<code>PB Node A1 / {farmId}-A1</code>	Block 1
<code>PB Node A2 / {farmId}-A2</code>	Block 2
<code>PB Node A3 ... A8</code>	Blocks 3 ... 8

Metric cards use:

- **Green** — live field sensor data (temperature, humidity, leaf wetness / soil moisture)
- **Blue** — local weather API for the remaining metrics

Blocks without a reporting node show weather-only data.

Staff admin access

Set `ADMIN_BOOTSTRAP_EMAILS` (comma-separated) in Railway. Listed addresses become staff on next sign-in.

Staff see an **Admin** tab to:

- Provision new customer farms
- View node online/offline status
- Issue or revoke access codes

- Remove farm members
- Rotate a node's ingest key

Admin API routes return **404** to non-staff (invisible, not just locked).

To grant admin manually:

```
UPDATE users SET is_admin = TRUE WHERE LOWER(email) = 'you@example.com';
```

After the customer redeems a code

First redeemer joins as **member**. To let them manage access codes and crew:

```
UPDATE farm_members SET role = 'owner'  
WHERE farm_id = '<farmId>' AND user_id = '<userId>';
```

Issuing a replacement access code

If the card is lost (the code cannot be recovered from the database):

```
npm run provision -- --farm-id farm_abc123 --add-code --label "replacement card"
```

Or use **Admin** → **farm detail** → **Issue code** in the app.

Rotating a node's ingest key

If a device is lost, compromised, or reflashed without the original key:

- **Admin tab** → farm → select node → **Rotate key**
- Or **POST** `/v1/admin/devices/:deviceId/rotate-key` (staff only)

The new key is shown once. Reflash the node with the new key.

Troubleshooting: customer can't see devices

Check	Requirement
Farm provisioned?	Admin or CLI ran with correct <code>--nodes</code> count
Access code redeemed?	Customer entered code after sign-in
Correct account?	Same email they used to claim the code
Node reporting?	Device uses matching <code>deviceId</code> + <code>x-device-key</code>
Recent data?	<code>last_seen_at</code> within 7 days (API filter)
Same farm?	Device <code>farm_id</code> matches the farm from the access code

Common mistakes

Mistake	Result
Using old ids like <code>pb-node-A1</code> without provisioning	Device belongs to legacy farm, not customer's
Using shared <code>SENSOR_INGEST_API_KEY</code> for new customers	Legacy path only; won't attach to new farm
Two access codes for two devices	Unnecessary — one code covers the whole farm
Customer signed in but skipped code entry	Stuck on access-code screen or empty sensors
Lost provisioning output	Must reissue code or rotate keys — cannot recover plaintext

Verify in the database

Latest reading per device:

```
SELECT DISTINCT ON (d.id, sr.sensor_type)
  d.id, d.name, d.farm_id, sr.sensor_type, sr.value, sr.unit,
  to_timestamp(sr.created_at / 1000.0) AT TIME ZONE 'America/New_York' AS sent_at
FROM sensor_readings sr
JOIN devices d ON d.id = sr.device_id
ORDER BY d.id, sr.sensor_type, sr.created_at DESC;
```

Who has access to a farm:

```
SELECT u.email, u.username, m.role, m.joined_at
FROM farm_members m
JOIN users u ON u.id = m.user_id
WHERE m.farm_id = '<farmId>';
```

Active access codes for a farm:

```
SELECT id, label, use_count, max_uses, expires_at, revoked_at
FROM farm_access_codes
WHERE farm_id = '<farmId>'
ORDER BY created_at DESC;
```

Pre-shipment checklist

- ☐ Provision farm with correct name and node count
- ☐ Save access code and all ingest keys (shown once)
- ☐ Print access code on card for the box
- ☐ Flash node 1 with `{farmId}-A1` + ingest key 1
- ☐ Flash node 2 with `{farmId}-A2` + ingest key 2
- ☐ (Repeat for A3...A{N} if more nodes)
- ☐ Verify each node reports (check Admin tab or SQL)
- ☐ Ship hardware + access code card (no ingest keys in the box)
- ☐ Tell customer to sign up, enter code, open Sensors tab

Quick reference

Question	Answer
How many access codes per farm?	One (covers all nodes)
How to name devices?	Auto: <code>{farmId}-A1</code> , <code>PB Node A1</code> , <code>Block 1</code>
Which code goes to the customer?	The <code>PB-XXXX-...</code> from provisioning
When do they see sensor data?	After redeeming code and nodes reporting
Where is tenancy enforced?	<code>farm_members</code> (users) + <code>devices.farm_id</code> (nodes)